Krishi Mitra AI

Kisan Context Intelligence Engine

Right farmer · Right moment · Right channel

Team **Elemental** · SYNGENTA × IITM BS · Track 1: AI-Powered Agricultural Marketing at Scale

The Problem: Generic, Untimely Outreach

Low Trust

Farmers receive generic SMS and untimely product pushes — low conversion, wasted spend.

No Context

Campaigns rarely combine who the farmer is (crop, district, language, device) with why today matters (weather, pest pressure, crop stage).

Fatigue

Same grower contacted too often without agronomic reason — fatigue builds, brand trust erodes.

Our Solution: A Decision Engine First

Farmer Data

Profile, weather, and crop calendar collected.

Unified Context

Combine signals into a single farmer view.

Urgency Check

Assess urgency and farmer fatigue before contact.

Product Rank

Recommend product and explain why now.

If urgency is low or fatigue is high, we suppress the touch — protecting brand trust. Every contact is **justified, timed, and actionable.**

What We Delivered — Round 1

1. **Farmer Database**
   Regional demo growers across Gujarat, Maharashtra, Bihar, Tamil Nadu, and Uttar Pradesh.

2. **Context Assembly**
   Crop stage, pest signals, humidity, and language unified in one API response.

3. **Urgency Engine**
   Explainable scores with channel recommendation — send, delay, or suppress.

4. **Product Ranker**
   12 Syngenta catalog products ranked with transparent match reasons.

5. **Multichannel Delivery**
   SMS, WhatsApp, and live AI agentic voice calls with personalized regional scripts.

FarmerContext — The Spine


```json
{
  "farmer_id": "GJ-014",
  "name": "Mayur",
  "district": "Anand, Gujarat",
  "crop": "cotton",
  "stage": "flowering",
  "pest_risk": "whitefly",
  "language": "Gujarati",
  "device": "feature_phone"
}
```


One JSON Object Per Farmer Per Run

The single source of truth for all downstream decisions.

- **Profile:** state, district, village, crops, farm size, language, device
- **Signals:** active pest, crop stage, weather anomaly, seasonal window

Also: Yashvi (MH-021) — Akola, Maharashtra · early kharif cotton · thrips/jassids · Marathi

Intelligence Layer: Urgency of communication

URGENCY	ENGAGEMENT	DEVICE	→ CHANNEL
HIGH ≥0.7	LOW <0.3	Keypad	🚜 Field Visit
HIGH ≥0.7	LOW <0.3	Smartphone	📞 Voice Call
HIGH ≥0.7	MED+ ≥0.3	Any	📱 WhatsApp
MED ≥0.4	Any	Smartphone	📱 WhatsApp
MED ≥0.4	Any	Keypad	💬 SMS
LOW <0.4	Any	Any	📱 WhatsApp / 💬 SMS

Urgency Scoring

Blends agronomic need (pest, stage, weather) with engagement fatigue. 3-layer hybrid engine that decides:

- Which farmer to contact
- When to contact
- How to contact
- Whether to suppress outreach

🌿 Deterministic Rules

urgency = 0.4 × pest_risk + 0.3 × weather_anomaly + 0.2 × crop_vulnerability+ 0.1 × (1 − recency_penalty)

🤖 ML Personalization

Engagement prediction using XGBoost

Known Farmers

XGBoost + CalibratedClassifierCV on clicked_status Model

Outputs a **Engagement Probability**

- District CTR
- Device type
- Farm size
- Recency
- Seasonality
- Product scan history

New Farmers

Population Priors

Blended: `0.5×district + 0.3×crop + 0.2×device`

- District average CTR: *Farmers in Nashik click at 8%*
- Crop average response rate: *Wheat farmers respond at 5.3%*
- Device-level engagement: *Smartphone users engage at 7%*

```
intervention_priority =
(1 − eng_weight) × urgency
+ eng_weight × engagement
```

```
expected_value =
(success_probability × benefit)
− channel_cost
```

Urgency weights: LogisticRegression on clicked_status + product_scan → normalized absolute coefficients

Messaged ≤3 days ago → **Hard suppress**

Messaged ≤7 days ago AND intervention_priority < 0.45 → **Soft suppress**

Critical bypasses (urgent messages still get through)

Intelligence Layer: Product Ranking

Scenario A: Wheat rust, high urgency

Crop: wheat | Pest: rust | Urgency: 0.9
History: Used Tilt (FRAC-3) last week

Result:
#1: Amistar 250 SC (FRAC-11) — different MoA ✓
#2: Movondo (FRAC-3+5) — rescue capability ✓
Score 250 EC rejected: same MoA as Tilt

Scenario C: Near harvest, fungal issue

Crop: wheat | Pest: fungal | Days to harvest: 7

Result:
All foliar sprays heavily penalized (PHI concern).
System may recommend no intervention or flag for field visit.
"Too close to harvest for foliar application"

Product Ranking

Given this farmer's crop, pest problem, urgency, and history — which 2 products should we recommend, and why?

```
efficacy = 0.25×crop_match + 0.30×pest_match + 0.15×stage_match + 0.20×efficacy_rating + 0.10×intent_alignment
```

Score 250 EC: Same MoA group (FRAC-3) used recently: rotate to prevent resistance
Movondo: Premium pricing — may not suit small farm budget
Cruiser 350 FS: Seed treatment not applicable at flowering growth stage
Kavach 75 WP: Low stock availability in your district

Multilingual Outreach

SMS

Concise advisory in farmer's language — Gujarati, Marathi, Bhojpuri, Tamil, Hindi.

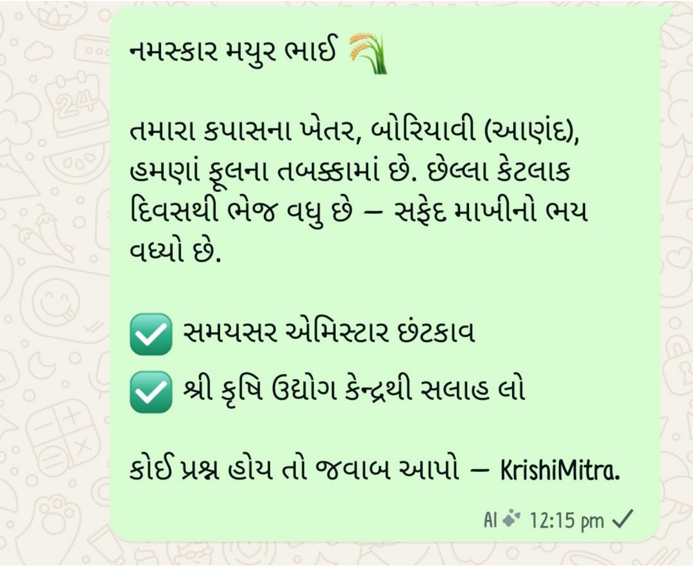
WhatsApp

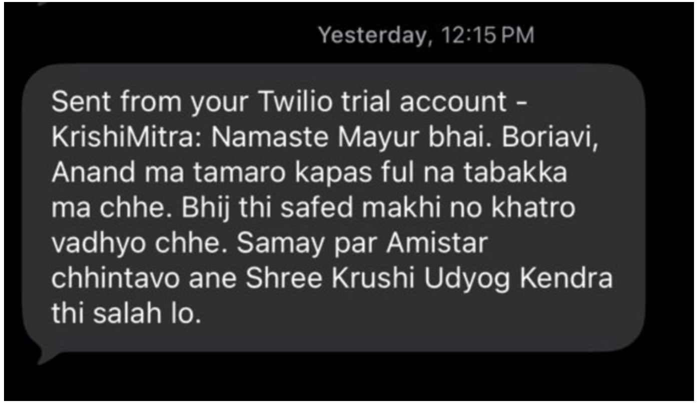
Structured message with crop, risk, product, and local retailer.

AI Agentic Voice Call

Personalized spoken opening, then live two-way conversation — built for feature phones and low-literacy users.

Live demos: **Mayur** (Gujarat cotton), **Yashvi** (Maharashtra cotton), **Rajnish** (Bihar rice nursery).





System Design

Modular microservices orchestrated for reproducible demos — one command via `run_all.sh`.

Microservices

Farmer profiles, weather, crop calendar, context builder, urgency, product ranking, campaign receptivity.

External Signals

Open-Meteo forecasts, ag-market and government ag-advisory datasets.

Channel Abstraction

SMS, WhatsApp, and telephony behind farmer JSON — swap providers without rewriting logic.

Demo Walkthrough · Impact & Differentiation

01

Start all services

Confirm health endpoints.

02

Fetch Mayur (GJ-014)

View Gujarat cotton context and whitefly advisory.

03

Call product ranking API

See top Syngenta match and reason codes.

04

Preview & send Gujarati SMS

Optional live send.

05

Place AI agentic voice call

Personalized Gujarati intro, then interactive Q&A.

Per-Farmer Context

Not one message for millions.

Send Only When Urgent

Suppress when fatigued.

Explained Product Advice

Ranked, not pushed blindly.

Device-Respectful Channels

SMS vs WhatsApp vs voice.

Roadmap & Team Elemental

What's Next

- → Deeper live weather → automated pest & disease risk models
- → Bilingual "why now" copy generation at scale
- → Batch campaign orchestrator with audit trail
- → Judge & ops dashboard for campaign analytics
- → Production sync with Syngenta India product master

Team Elemental

- **Yashvi Upadhyay** — Backend & APIs
- **Rajnish Kumar** — Outreach channels, AI voice, architecture
- **Mayur H. Doshi** — ML: urgency scorer, product ranker
- **Agrim Srivastava** — Database schema & seed data
- **Tarang Jhaveri** — ML & agronomic research

Source code on GitHub